Exploratory Data Analysis (EDA) Report

E-Commerce Sales Dataset — Orders, Products, Revenue & Customer Behaviour

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Program	BS Computer Science (Session 2024–2028)
Focus Area	Data Analytics
Dataset Size	1,200 Orders | 17 Attributes
Report Type	Exploratory Data Analysis (EDA)

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1. Introduction & Objectives

This report presents an Exploratory Data Analysis (EDA) of an e-commerce sales dataset containing 1,200 customer orders recorded between 2023 and 2025. The purpose of this analysis is to understand the underlying structure of the data, identify patterns in sales, products, payment behaviour, and customer acquisition channels, and to surface actionable business insights that can support data-driven decision-making.

Objectives of this analysis:

- Understand the structure, quality, and completeness of the dataset.
- Summarize key numerical variables using descriptive statistics.
- Identify sales trends across time (monthly and yearly).
- Analyze product performance, payment preferences, and referral channels.
- Detect outliers and unusual patterns in order values.
- Examine relationships between variables using correlation analysis.
- Translate findings into clear, actionable business insights.

2. Dataset Overview

The dataset consists of 1,200 order-level records with 17 attributes covering order identifiers, product details, pricing, shipping, payment, coupon usage, and referral source. The table below summarizes the key fields used in this analysis.

Column	Description	Type
OrderID	Unique identifier for each order	Text
Date	Date the order was placed	Date
Product	Product purchased (7 categories)	Categorical
Quantity	Number of units purchased	Numeric
UnitPrice	Price per unit ($)	Numeric
TotalPrice	Total order value ($)	Numeric
PaymentMethod	Payment channel used (5 types)	Categorical
OrderStatus	Delivered / Shipped / Cancelled / Returned / Pending	Categorical
CouponCode	Coupon applied, if any	Categorical
ReferralSource	Marketing channel that drove the order	Categorical
ShippingAddress	Delivery address (655 unique)	Text

No missing (null) values were found in any of the 17 columns across all 1,200 records, indicating the dataset was already cleaned prior to this analysis.

3. Data Cleaning & Preparation

Before analysis, the raw order data was processed into a 'Cleaned_Data' sheet with derived time features (Month, Year, Month-Year) to support trend analysis. The cleaning process confirmed:

• Zero null values across all 17 columns for all 1,200 orders.

• Consistent data types: dates parsed correctly, prices stored as numeric (float), quantities as integers.

• Categorical fields (Product, PaymentMethod, OrderStatus, CouponCode, ReferralSource) standardized into fixed sets of categories.

• Derived fields — Month, Year, and Month_Year — were engineered from the Date column to enable time-series analysis.

• 8 statistical outliers were identified in TotalPrice using the IQR method (see Section 9) and retained as legitimate high-value orders rather than removed as errors.

4. Descriptive Statistics

Summary statistics were computed for the four core numeric variables: TotalPrice, UnitPrice, Quantity, and ItemsInCart. These statistics describe the central tendency, spread, and shape of each distribution.

Statistic	TotalPrice ($)	UnitPrice ($)	Quantity	ItemsInCart
Mean	1,053.97	356.41	2.95	5.49
Median	823.62	364.21	3.00	5.00
Std. Deviation	819.86	197.18	1.41	2.28
Minimum	11.39	11.39	1	1
Maximum	3,456.40	699.93	5	10
Skewness	0.89 (right-skewed)	-0.03 (~symmetric)	0.03 (~symmetric)	0.00 (symmetric)
Count	1,200	1,200	1,200	1,200

TotalPrice shows a right-skewed distribution (skewness = 0.89) — most orders cluster at lower values, with a long tail of higher-value transactions pulling the mean ($1,053.97) above the median ($823.62). UnitPrice and Quantity are both close to symmetric, suggesting products are priced and purchased fairly uniformly across the catalogue.

1,200	$1.26M	$1,054	655
Total Orders	Total Revenue	Avg Order Value	Ship Locations

5. Univariate Analysis

This section examines individual variables in isolation to understand their distribution across the dataset — starting with product performance, payment methods, and order status.

5.1 Revenue by Product

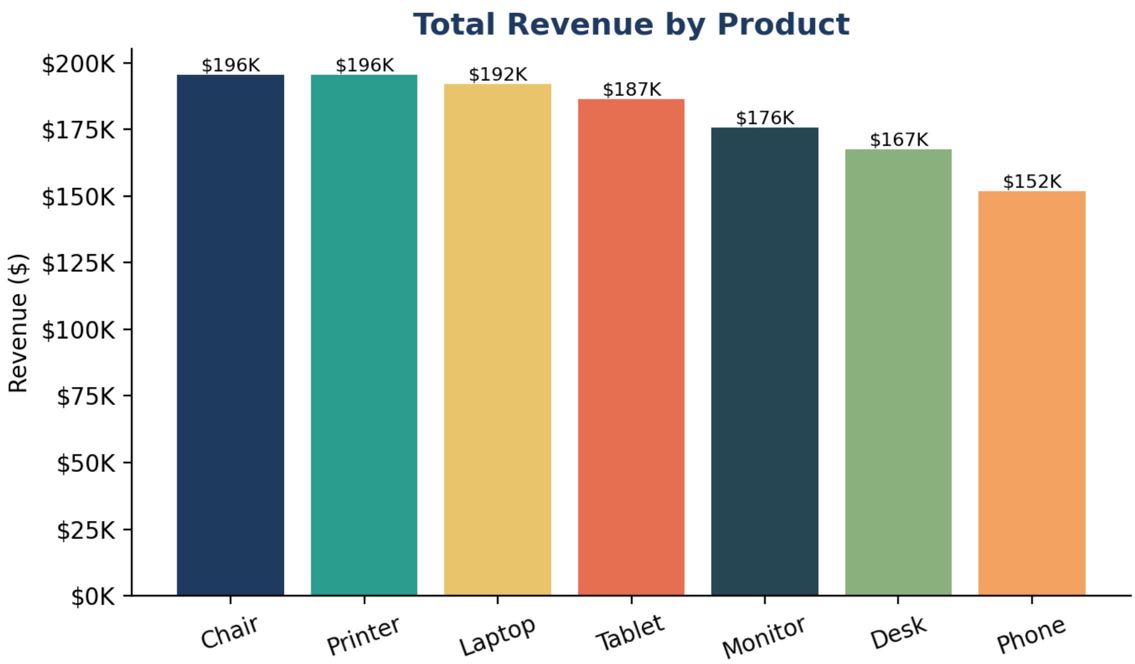

Figure 1: Total revenue generated by each product category.

Chair leads all products in total revenue ($195,620) and units sold (562), narrowly ahead of Printer ($195,613, 542 units). Phone generated the lowest revenue ($151,722), suggesting either lower pricing or weaker demand relative to other categories.

5.2 Payment Method Distribution

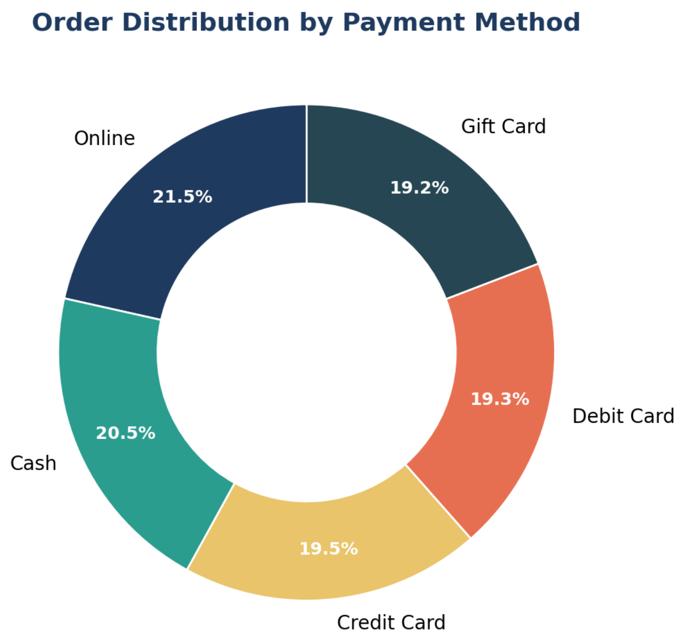

Figure 2: Share of orders by payment method.

Online payments are the most-used method (21.5% of orders), but usage is fairly evenly spread across all five payment channels (Cash, Credit Card, Debit Card, Gift Card, Online), each falling between 19% and 21.5% — indicating no single dominant payment preference among customers.

5.3 Order Status Distribution

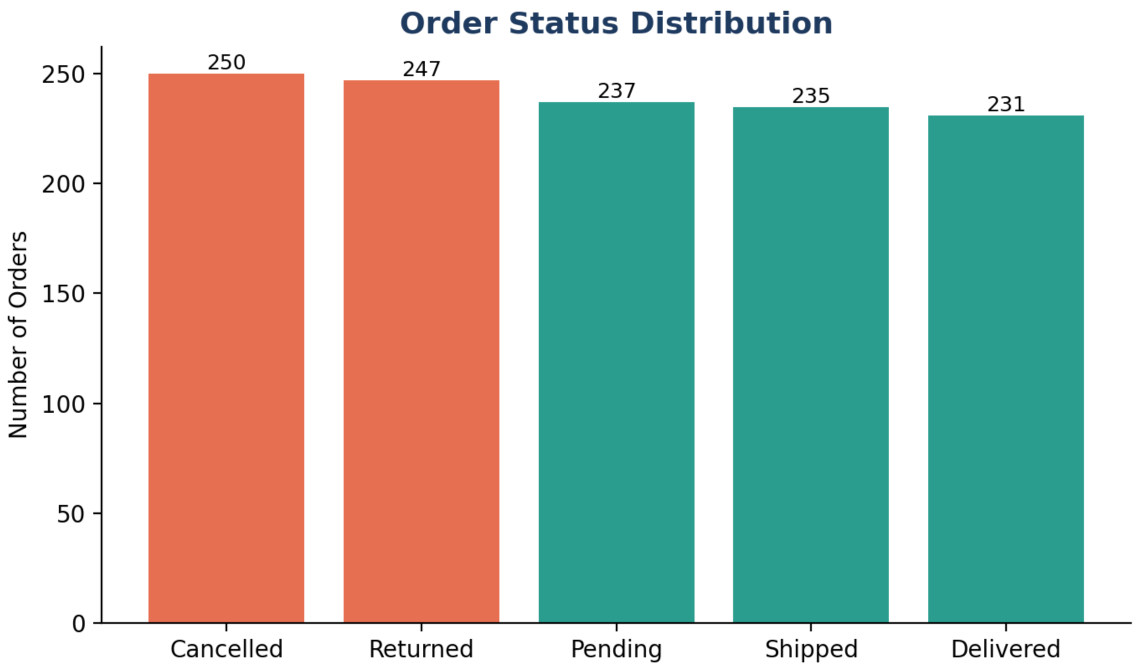

Figure 3: Number of orders by fulfillment status (red = unfulfilled, green = fulfilled).

Cancelled (250) and Returned (247) orders together account for 41.4% of all orders — notably higher than Delivered orders alone (231, 19.3%). This is a significant fulfillment concern that merits closer investigation into causes (e.g., payment failures, delivery delays, product quality, or customer dissatisfaction).

5.4 Referral Source Distribution

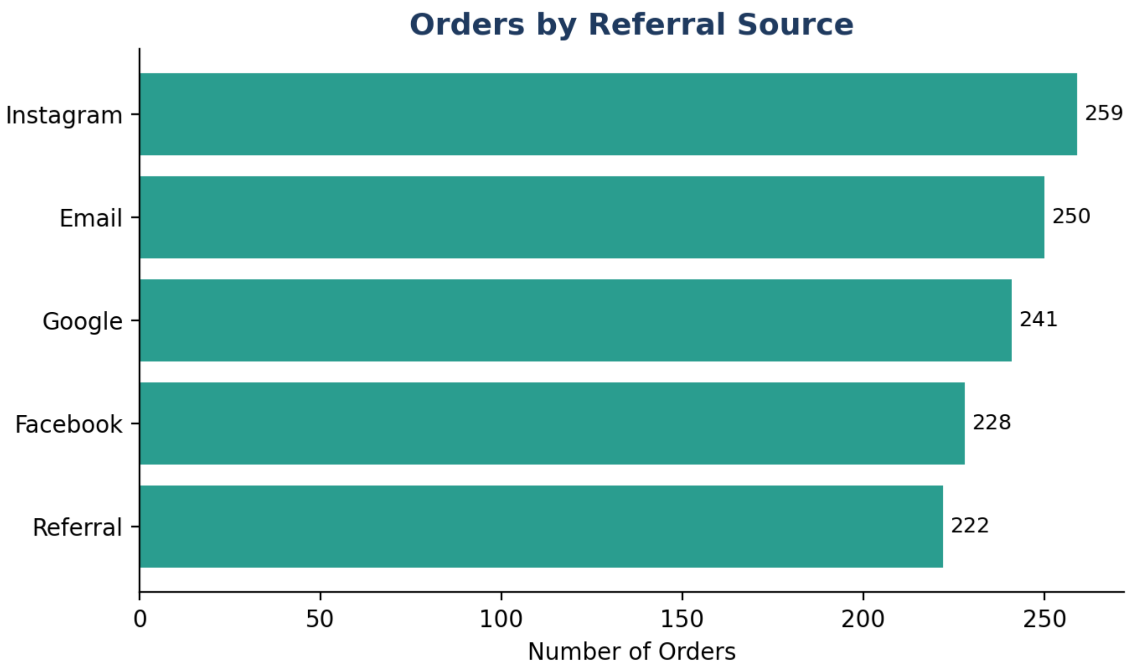

Figure 4: Number of orders attributed to each marketing/referral channel.

Instagram is the most successful referral source, driving 259 orders (21.6%) and $275,285 in revenue — ahead of Email, Google, Facebook, and direct Referral. This suggests social media, and Instagram specifically, is currently the strongest customer acquisition channel.

6. Bivariate & Correlation Analysis

To understand how numeric variables relate to one another, a Pearson correlation matrix was computed across Quantity, UnitPrice, TotalPrice, and ItemsInCart.

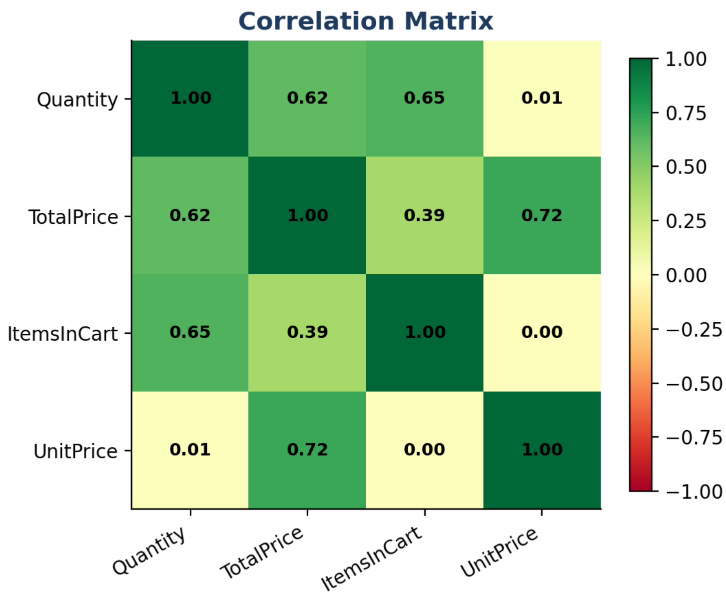

Figure 5: Correlation matrix of key numeric variables.

Variable Pair	Correlation (r)	Interpretation
UnitPrice ↔ TotalPrice	0.72	Strong positive — price is the primary driver of order value
Quantity ↔ TotalPrice	0.62	Moderate positive — quantity secondarily drives order value
Quantity ↔ ItemsInCart	0.65	Moderate positive — larger carts tend to have higher quantities
UnitPrice ↔ Quantity	0.01	Negligible — price does not influence units purchased

Order value (TotalPrice) is driven primarily by unit price (r = 0.72) and secondarily by quantity purchased (r = 0.62). Notably, UnitPrice and Quantity are essentially uncorrelated (r = 0.01), meaning customers do not systematically buy more units of cheaper products — purchase quantity appears to be an independent decision from price sensitivity.

7. Sales & Revenue Trends

7.1 Monthly Sales Trend

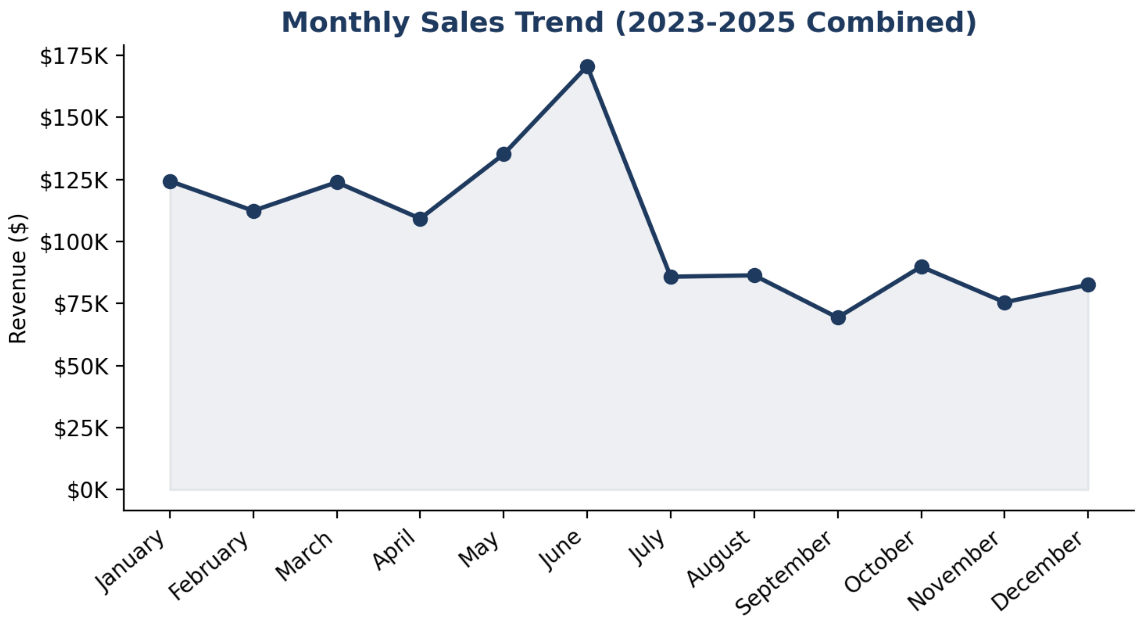

Figure 6: Combined monthly revenue across 2023-2025.

June recorded the highest sales month overall ($68,069 in single-year terms, and the strongest month when aggregated), while activity tapers off in the second half of the year, with September and November being comparatively weaker. April also shows a seasonal dip relative to neighbouring months.

7.2 Yearly Revenue Trend

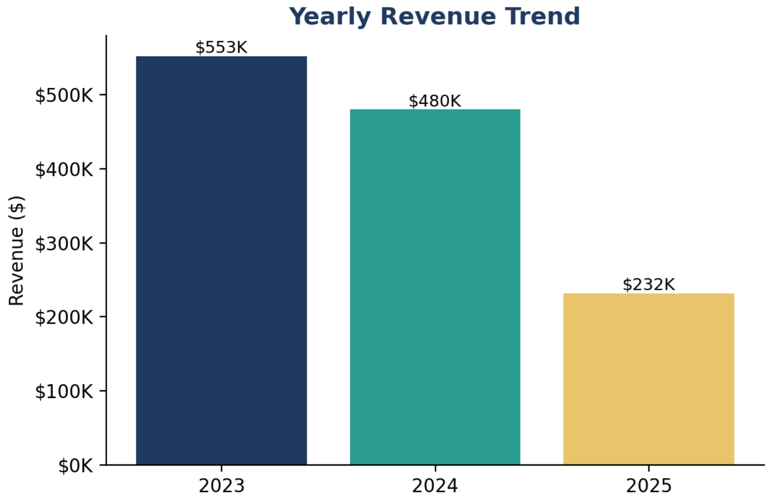

Figure 7: Total revenue by year.

Revenue declined year-over-year across the dataset: $552,643 (2023) → $480,236 (2024) → $231,883 (2025, partial year). Since 2025 data is likely incomplete (partial-year coverage), this decline should be interpreted with caution — it may reflect the reporting window rather than genuine business decline. A like-for-like comparison using only completed months is recommended before drawing firm conclusions.

7.3 Coupon Effectiveness

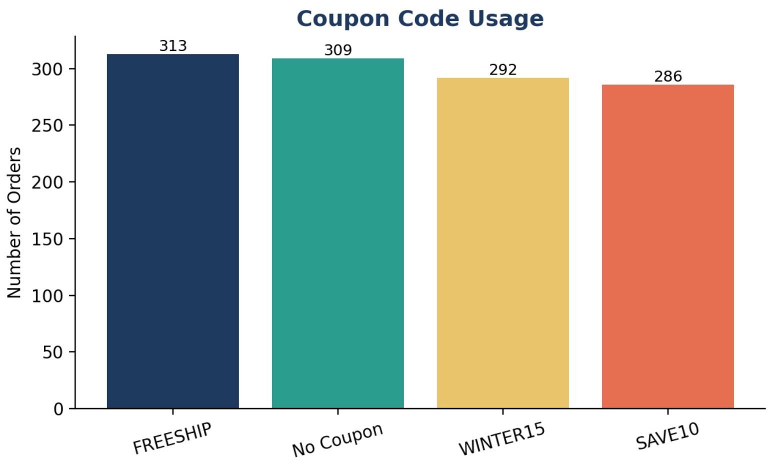

Figure 8: Number of orders by coupon code applied.

FREESHIP is the most redeemed coupon, used in 26.1% of orders (313 orders), and is associated with the highest overall revenue among coupon types — suggesting customers respond strongly to free-shipping incentives compared to percentage-based discounts (SAVE10, WINTER15).

8. Customer Behaviour Analysis

Order value analysis (Section 4) shows an average order value of $1,053.97 against a median of $823.62 — the gap confirms a right-skewed spending pattern where a smaller group of high-value orders raises the average above the typical (median) purchase.

Shipping data spans 655 unique addresses with no single location accounting for more than 0.6% of total revenue (the top address, 533 Main St, had only 8 orders) — indicating a geographically dispersed customer base with no heavy concentration risk in any single area.

Across payment methods, spending is broadly balanced: Credit Card generated the highest revenue ($263,848) despite Online having more orders (258 vs 234), implying Credit Card customers tend to place higher-value orders on average.

9. Outlier Analysis

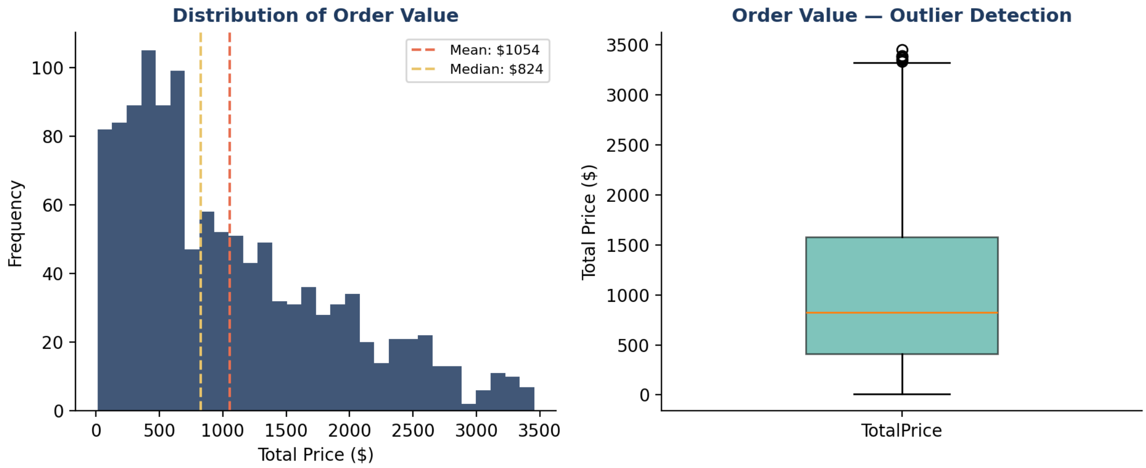

Figure 9: Distribution (left) and boxplot (right) of order value, highlighting the right-skew and outliers.

Using the Interquartile Range (IQR) method, 8 orders were identified as statistical outliers exceeding the upper bound for TotalPrice. On investigation, these appear to be legitimate bulk or high-value purchases (large quantity × high unit price combinations) rather than data entry errors, and were therefore retained in the dataset rather than removed.

10. Key Business Insights

#	Insight
1	Chair is the best-selling product by both units (562) and revenue ($195,620), narrowly ahead of Printer.
2	Average order value is $1,053.97 (median $823.62) — a right-skewed distribution with a tail of large orders.
3	Online payment is the most-used method (21.5%), though usage is fairly even across all five methods.
4	Instagram is the top referral source by both order volume (259) and revenue ($275,285).
5	June is the peak sales month; April tends to be seasonally weaker.
6	Orders span 655 shipping addresses with no location driving more than 0.6% of revenue.
7	8 orders exceed the IQR outlier bound — legitimate bulk/high-value purchases, not errors.
8	FREESHIP is the most redeemed coupon (26.1% of orders) and drives the highest associated revenue.
9	Order value is driven primarily by unit price (r=0.72), secondarily by quantity (r=0.62).
10	Cancelled + Returned orders make up 41.4% of all orders — well above Delivered alone (19.3%).

11. Conclusion & Recommendations

The exploratory analysis of this 1,200-order e-commerce dataset reveals a business with a diversified, well-balanced product and payment mix, but one facing a significant fulfillment challenge. The following recommendations follow directly from the findings above:

- **Investigate cancellations and returns:** With 41.4% of orders Cancelled or Returned, root-cause analysis (payment failures, delivery delays, product quality) should be prioritized to protect revenue and customer trust.
- **Double down on Instagram-driven acquisition:** As the top-performing referral channel by both volume and revenue, increased investment here is likely to yield strong returns.
- **Promote free-shipping incentives:** FREESHIP outperforms percentage discounts in redemption and revenue impact, suggesting shipping cost is a stronger purchase driver than price discounts for this customer base.
- **Plan inventory around seasonal peaks:** June consistently outperforms other months; ensure stock and staffing are aligned with this seasonal pattern, particularly for top products (Chair, Printer, Laptop).
- **Treat high-value outlier orders as a distinct segment:** The 8 bulk/high-value orders identified should be studied separately, as they may represent a B2B or wholesale customer segment worth targeting directly.
- **Re-verify the 2025 year-over-year decline:** Since 2025 appears to be a partial year, revenue comparisons should be normalized to completed months before concluding a genuine downward trend.

Report prepared by Muhammad Adnan Arif, BS Computer Science, University of Layyah, as part of Exploratory Data Analysis (EDA) coursework/project work.